

The Origin of the High-resolution NMR Spectrum Mirror Symmetry of the $AA'BB'$ ($AA'XX'$) Spin System

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Abstract

A comprehensive theoretical analysis of the origin of NMR spectrum mirror symmetry of the $AA'BB'$ ($AA'XX'$) spin system was performed in the framework of the action of the generalized parity operator $\hat{Q} = \hat{P}\hat{\Pi}$. We demonstrate that despite the lack of geometric bisymmetry in the Hamiltonian matrix (due to $J_{AA'} \neq J_{BB'}$), the system exhibits a hidden dynamic symmetry. It is shown that the spectral invariance arises from the specific algebraic structure of the Hamiltonian, which guarantees the invariance of the characteristic polynomial under parameter exchange.

1 Introduction

This work analyzes the origin of the spectral mirror symmetry in C_2 -symmetric $AA'BB'$ spin systems in relation to the structure of the Hamiltonian matrix representations. The total spin Hamiltonian is given by $\hat{H} = \hat{H}_Z + \hat{H}_{SS}$. As we have shown previously, the mirror symmetry of the spectrum is determined by the nature of the generalized parity operator $\hat{Q}$ action on the spin Hamiltonian $\hat{H}$. Since the spin system is characterized by unequal intra-group coupling constants ($J_{AA'} \neq J_{BB'}$), the Hamiltonian does not commute with this operator ($[\hat{H}, \hat{Q}] \neq 0$). Therefore, we examine the effect of the generalized parity operator $\hat{Q}$ on the Hamiltonian and its matrix representations. We reveal that the condition $[\hat{H}, \hat{Q}] = 0$ is sufficient but not necessary for spectral symmetry.

2 Representations in P-Basis and Q-Basis

2.1 The P-Basis (Product Basis)

The P-basis is partitioned into subspaces according to the total angular momentum projection M_z . Crucially, the ordering is defined such that the entire basis sequence forms a palindrome with respect to the collective spin inversion operator $\hat{P}$. This means the second half of the basis is the state-by-state spin inversion of the first half (in reverse order).

- **Zeeman Hamiltonian ($\hat{H}_Z$):** The matrix is diagonal. Since the spectrum is centered, the elements exhibit *anti-persymmetry*:

$$(H_Z)_{i,i} = -(H_Z)_{N-i+1, N-i+1}$$

The values on the main diagonal are inverted with respect to the anti-diagonal (e.g., $+\nu_A - \nu_B/2$ at the top, $-\nu_A - \nu_B/2$ at the bottom).

- **Spin-Spin Hamiltonian ($\hat{H}_{SS}$):** The matrix is *bisymmetric* (symmetric with respect to both the main and anti-diagonals). This structure is a direct mathematical consequence of the palindromic P-ordering described above and the invariance of the scalar product $\mathbf{I}_i \cdot \mathbf{I}_j$ under collective spin inversion.

2.2 The Q-Basis (Generalized Parity Basis)

The Q-basis is constructed to align with the generalized parity operator $\hat{Q} = \hat{P}\hat{\Pi}$. To understand the matrix structure, we must distinguish between sectors where the Q-basis differs from the P-basis and where it coincides.

2.2.1 The Outer Shells ($M_z \neq 0$)

In these subblocks, the Q-basis is constructed by reversing the spin order of the P-basis states in the second half of the sequence. This creates mirror-image blocks (e.g., $M_z = +1$ and $M_z = -1$).

- **Structure of $\hat{H}_{SS}$:** The global bisymmetry is lost. The "lower" block ($M_z < 0$) is the geometric reflection of the "upper" block ($M_z > 0$), but with a **complete parameter substitution** $J_{AA'} \leftrightarrow J_{BB'}$. Since $J_{AA'} \neq J_{BB'}$, the reflection symmetry relative to the anti-diagonal is broken.
- **Structure of $\hat{H}_Z$:** The matrix is diagonal and exhibits global *persymmetry*. This is a direct consequence of the resonance frequency balance ($\nu_A = -\nu_B$): the spin inversion inherent in the Q-basis is compensated by the spatial permutation of frequencies, resulting in equal energies at symmetric positions ($H_{ii} = H_{N-i+1, N-i+1}$).

2.2.2 The Zero-Quantum Block ($M_z = 0$)

In this central subblock, the Q-basis identifies and isolates a special structure: the *self-conjugate Core*. This Core consists of specific state combinations, such as the triplet-triplet pairs (T_+T_- , T_-T_+) and the two central product states ($\alpha\beta\alpha\beta$ and $\beta\alpha\beta\alpha$), which map onto themselves under the action of $\hat{Q}$.

We group these core states in the center of the matrix, maintaining the standard P-ordering within the group. Crucially, the ordering of the remaining "outer" product states in this block (the first state $\alpha\beta\beta\alpha$ and the last state $\beta\alpha\alpha\beta$) coincides in both the P-basis and the Q-basis. Therefore, effectively, the state ordering throughout the entire zero-quantum block corresponds to the P-basis.

1. **Inner Shells:** These are the paired states surrounding the core that map into each other under $\hat{Q}$.
 - *Spin-Spin Term:* The matrix retains *bisymmetry* because the basis coincides with the P-basis. However, unlike the isolated outer shells, these states are strongly mixed with the Core components. This creates a symmetry conflict: the "shell-like" tendency to manifest parameter asymmetry ($J_{AA'} \neq J_{BB'}$) is constrained by the rigid, bisymmetric structure of the dense central block.
 - *Zeeman Term:* Exhibits *persymmetry*. Even though the basis is P-ordered, the specific pairing of these shell states results in symmetric energy distribution ($H_{ii} = H_{N-i+1, N-i+1}$), similar to the outer shells.
2. **The Core:** These are the self-conjugate states grouped in the center.
 - *Spin-Spin Term:* Retains *bisymmetry*, forming a rigid, invariant backbone for the central block.
 - *Zeeman Term:* Retains *anti-persymmetry* ($H_{ii} = -H_{N-i+1, N-i+1}$), mirroring the standard behavior of the P-basis diagonal.

3 Conjugation by Operator $\hat{Q}$

The action of the operator $\hat{Q}$ on the Hamiltonian corresponds to the transformation $\hat{H}' = \hat{Q}^\dagger \hat{H} \hat{Q}$. The manifestation of this transformation depends critically on the chosen representation basis.

3.1 Action in the P-Basis

In the Product Basis, the spin-spin Hamiltonian matrix is naturally bisymmetric. Consequently, the geometric aspect of the transformation (reflection relative to the anti-diagonal) maps the matrix onto itself. The transformation manifests purely algebraically:

- **Zeeman Term:** Remains invariant due to the centered frequency balance ($\nu_A = -\nu_B$).
- **Spin-Spin Term:** The matrix structure remains identical, but the coupling constants are swapped ($J_{AA'} \leftrightarrow J_{BB'}$). Since $J_{AA'} \neq J_{BB'}$, the numerical values change, but the geometric symmetry (bisymmetry) is preserved.

3.2 Action in the Q-Basis

In the Q-Basis, the transformation splits into distinct phenomena depending on the topological sector:

1. **Outer Shells ($M_z \neq 0$): Global Reflection** For the spatially separated blocks, the transformation acts as a geometric reflection across the anti-diagonal. The upper shell is mapped to the physical position of the lower shell. This acts as a **mutual reflection** of the blocks relative to the matrix center, physically relocating the parameter sets.
2. **Zero-Quantum Block ($M_z = 0$):** The behavior of this Hamiltonian block under $\hat{Q}$ is identical to the transformation observed in the P-basis. The matrix is geometrically invariant (mapped onto itself), and the transformation is purely algebraic.
 - **Inner Shells:**
 - *Spin-Spin Term:* Manifests as an *in-place parameter swap* ($J_{AA'} \leftrightarrow J_{BB'}$). Since these states are mixed, the numerical values change, creating a conflict with the geometric invariance.
 - *Zeeman Term:* Exhibits persymmetry and remains invariant under the transformation.
 - **The Core:**
 - *Spin-Spin Term:* Is bisymmetric and maps onto itself. Numerical values remain strictly invariant because the matrix elements in this sector depend exclusively on the symmetric parameter sums (ΣJ_{intra}), and the asymmetry parameter ΔJ_{intra} is structurally absent.
 - *Zeeman Term:* Exhibits anti-persymmetry, which is structurally preserved (with consistent sign inversion) under the transformation.

4 Symmetrized Representation

Transitioning to a symmetry-adapted basis via the unitary transformation U block-diagonalizes the Hamiltonian into a symmetric subspace and an asymmetric subspace according to the irreducible representations of the C_2 -symmetry group of the spin system.

To simplify the mathematical expressions in the following sections and avoid fractional coefficients, we analyze the *doubled* Hamiltonian matrices ($2\hat{H}$). We also define the resonance frequency difference $\Delta = |\nu_A - \nu_B|$ and introduce the following aggregate notation for the coupling constants:

$$\begin{aligned}\Sigma J_{intra} &= J_{AA'} + J_{BB'}, & \Delta J_{intra} &= J_{AA'} - J_{BB'} \\ \Sigma J_{inter} &= J_{AB} + J_{AB'}, & \Delta J_{inter} &= J_{AB} - J_{AB'}\end{aligned}$$

4.1 The Symmetric Subspace

The symmetric subspace corresponds to states that transform according to the trivial representation of the permutation symmetry.

- **Invariance:** The Hamiltonian block $\hat{H}_{sym}$ in this subspace is strictly invariant under the action of the generalized parity operator:

$$[\hat{H}_{sym}, \hat{Q}] = 0 \quad \text{and} \quad \hat{H}_{sym} = \hat{Q}^\dagger \hat{H}_{sym} \hat{Q}$$

- **Geometric Symmetry:** Consequently, the Hamiltonian matrix retains global bisymmetry (symmetry with respect to both the main and secondary diagonals).

4.1.1 Structure of the Zero-Quantum Block

The central block of the symmetric subspace ($M_z = 0$) exhibits a complex spectral structure characterized by four distinct non-degenerate eigenvalues. The explicit form of this 4×4 doubled Hamiltonian block illustrates the extensive connectivity between basis states:

$$\begin{pmatrix} -\frac{1}{2}\Sigma J_{intra} - \Delta J_{inter} & \sqrt{2}J_{AB} & \sqrt{2}J_{AB} & \Sigma J_{intra} \\ \sqrt{2}J_{AB} & \frac{1}{2}\Sigma J_{intra} - \Sigma J_{inter} + 2\Delta & 0 & \sqrt{2}J_{AB'} \\ \sqrt{2}J_{AB} & 0 & \frac{1}{2}\Sigma J_{intra} - \Sigma J_{inter} - 2\Delta & \sqrt{2}J_{AB'} \\ \Sigma J_{intra} & \sqrt{2}J_{AB'} & \sqrt{2}J_{AB'} & -\frac{1}{2}\Sigma J_{intra} + \Delta J_{inter} \end{pmatrix} \quad (1)$$

Physically, this non-degeneracy arises from the strong mixing between the central product states ($\alpha\beta\alpha\beta, \beta\alpha\beta\alpha$) and the zero-quantum projections of the triplet-triplet states (vector products, $T_A T_B$). Unlike the asymmetric subspace, where the central block structure allows for symmetry protection, the symmetric Hamiltonian fully couples the scalar product states with the triplet-triplet components (notice the non-zero off-diagonal terms involving $\sqrt{2}J$ and the direct coupling ΣJ_{intra} in the corners). This hybridization lifts all degeneracies, resulting in a unique set of energy levels dependent on the interplay of all coupling constants.

Despite this complex internal mixing, the symmetry of the observable subspectrum is strictly preserved. The eigenstates of the zero-quantum block serve as central "pivots" for transitions. Since the energy levels of the $M_z = \pm 1$ blocks are symmetric counterparts, the set of transitions connecting them to the unique $M_z = 0$ central block forms a perfectly palindromic spectral pattern. Thus, the complex internal mixing within the central block does not compromise the spectral symmetry. Since the Hamiltonian in this subspace depends exclusively on parameter sums, it remains strictly invariant under the parameter exchange ($\hat{H}' = \hat{H}$), providing a *statically symmetric core* to the overall spectrum.

4.2 The Asymmetric Subspace

To illustrate the origin of isospectrality, we decompose the Hamiltonian matrix in the asymmetric subspace into four distinct components. This decomposition allows us to isolate the specific terms responsible for the symmetry breaking. The decomposition of the doubled Hamiltonian matrix $2\hat{H}_{asym}$ is given by:

$$\begin{aligned}
 2\hat{H}_{asym} = & \underbrace{-\frac{1}{2}\Sigma J_{intra}\mathbf{I}_6}_{\text{Scalar Shift}} + \underbrace{\begin{pmatrix} \Delta & 0 & 0 & 0 & 0 & 0 \\ 0 & -\Delta & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\Delta & 0 \\ 0 & 0 & 0 & 0 & 0 & \Delta \end{pmatrix}}_{\text{Zeeman Term}} + \underbrace{\begin{pmatrix} 0 & \Delta J_{inter} & 0 & 0 & 0 & 0 \\ \Delta J_{inter} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\Delta J_{inter} & 0 & 0 & 0 \\ 0 & 0 & 0 & \Delta J_{inter} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \Delta J_{inter} \\ 0 & 0 & 0 & 0 & \Delta J_{inter} & 0 \end{pmatrix}}_{\text{Invariant Coupling}} \\
 & + \underbrace{\begin{pmatrix} \Delta J_{intra} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\Delta J_{intra} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\Delta J_{intra} & 0 & 0 \\ 0 & 0 & -\Delta J_{intra} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \Delta J_{intra} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\Delta J_{intra} \end{pmatrix}}_{\text{Asymmetric Part}}
 \end{aligned} \tag{2}$$

This decomposition classifies the terms based on their behavior under the generalized parity operator $\hat{Q}$ (parameter exchange $J_{AA'} \leftrightarrow J_{BB'}$):

1. The **Scalar Shift** affects all eigenvalues equally and does not influence the spectral structure; we omit it in the subsequent analysis.
2. The **Zeeman Term** and the **Invariant Coupling Term** depend on parameters ($\nu, J_{AB}, J_{AB'}$) that are unaffected by the $A' \leftrightarrow B'$ permutation. Thus, they remain invariant under $\hat{Q}$.
3. The **Asymmetric Part** depends on ΔJ_{intra} . Since the parameter exchange swaps $J_{AA'}$ and $J_{BB'}$, this term changes its sign under $\hat{Q}$.

Combining the three significant terms, we obtain the effective doubled Hamiltonian $2\hat{H}_{asym}$. The action of $\hat{Q}$ flips the sign of the asymmetric part, resulting in the transformed matrix:

$$\underbrace{\begin{pmatrix} d_+ & \Delta J_{inter} & 0 & 0 & 0 & 0 \\ \Delta J_{inter} & -d_+ & 0 & 0 & 0 & 0 \\ 0 & 0 & -\Delta J_{inter} & -\Delta J_{intra} & 0 & 0 \\ 0 & 0 & -\Delta J_{intra} & \Delta J_{inter} & 0 & 0 \\ 0 & 0 & 0 & 0 & -d_- & \Delta J_{inter} \\ 0 & 0 & 0 & 0 & \Delta J_{inter} & d_- \end{pmatrix}}_{2\hat{H}_{asym}} \xrightarrow{\hat{Q}} \underbrace{\begin{pmatrix} d_- & \Delta J_{inter} & 0 & 0 & 0 & 0 \\ \Delta J_{inter} & -d_- & 0 & 0 & 0 & 0 \\ 0 & 0 & -\Delta J_{inter} & +\Delta J_{intra} & 0 & 0 \\ 0 & 0 & +\Delta J_{intra} & \Delta J_{inter} & 0 & 0 \\ 0 & 0 & 0 & 0 & -d_+ & \Delta J_{inter} \\ 0 & 0 & 0 & 0 & \Delta J_{inter} & d_+ \end{pmatrix}}_{2\hat{Q}^\dagger \hat{H}_{asym} \hat{Q}} \tag{3}$$

where $d_{\pm} = \Delta \pm \Delta J_{intra}$.

Under the action of $\hat{Q}$, the blocks $M_z = \pm 1$ reflect across the anti-diagonal, while in the block $M_z = 0$, a sign inversion of the off-diagonal elements occurs due to the swap $J_{AA'} \leftrightarrow J_{BB'}$.

4.2.1 Invariance of the Characteristic Polynomial

The structural changes involve a simultaneous sign flip of the diagonal Zeeman contribution and the coupling difference.

- **Zero-Quantum Block:** The asymmetric term ΔJ_{intra} appears only in the off-diagonal positions. Since the eigenvalues depend on the square of the off-diagonal term, the sign flip at ΔJ_{intra} leaves the block spectrum invariant.
- **High-Quantum Blocks:** The $\hat{Q}$ -transformation swaps the effective diagonal parameters d_+ and d_- . The diagonal matrix elements in both the $M_z = +1$ and $M_z = -1$ subblocks exhibit a simultaneous sign inversion of the ΔJ_{intra} and Zeeman (Δ) terms, allowing them to be parameterized as d_+ and d_- . Consequently, the characteristic polynomial depends on the product $d_+^2 \cdot d_-^2$ and remains invariant under the action of $\hat{Q}$ (or under the exchange $J_{AA'} \leftrightarrow J_{BB'}$). Thus, the eigenvalue spectrum of the outer blocks is preserved.

Since the contributions of both the central block (quadratic dependence) and the wings (permutation) are conserved, the characteristic polynomials of the initial and transformed matrices are identical:

$$P(\lambda) = \det(\hat{H}_{asym} - \lambda \mathbf{I}) = \det(\hat{Q}^\dagger \hat{H}_{asym} \hat{Q} - \lambda \mathbf{I})$$

This implies that the set of eigenvalues is invariant under the parameter exchange.

4.2.2 Dynamic Symmetry vs. Static Asymmetry

It is crucial to emphasize that the invariance of the characteristic polynomial, contrasting with the non-invariance of the Hamiltonian matrix itself, provides the formal evidence for *dynamic symmetry*.

While *static symmetry* would require the Hamiltonian to commute with the symmetry operator ($[\hat{H}, \hat{Q}] = 0$, implying $\hat{H} = \hat{Q}^\dagger \hat{H} \hat{Q}$), the asymmetric subspace behaves differently. Here, the Hamiltonian undergoes a structural transformation, yet the observable spectrum remains unchanged.

In this subspace, the energy levels are "crooked" (skewed) due to the non-zero difference ΔJ_{intra} , and do not form balanced pairs around the center. However, the specific algebraic structure ensures that these distortions are exactly compensated. The fact that $P(\lambda)$ remains invariant despite the breakdown of geometric symmetry confirms that the system possesses a hidden dynamic symmetry, resulting in a perfectly palindromic transition spectrum.

5 Conclusion

The analysis of the $AA'BB'$ system in the Q-representation reveals that the geometric asymmetry of the Hamiltonian matrix does not preclude spectral symmetry. By decomposing the Hamiltonian, we isolated the symmetry-breaking "Asymmetric Part" and demonstrated that its effect is either neutralized by quadratic dependence (in the zero-quantum block) or compensated by consistent sign inversions (in the high-quantum blocks).

This mechanism confirms the existence of *dynamic symmetry*: while the energy levels themselves may lack mirror symmetry due to unequal intragroup couplings ($J_{AA'} \neq J_{BB'}$), the observable transition spectrum remains perfectly palindromic. The system is thus isospectral under the parameter exchange, proving that spectral symmetry is a robust feature emerging from the specific algebraic structure of the Hamiltonian.